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Studierichting: Informatica
Oefeningenreeks 5A nr. 7.7

$$\int_0^{\frac{\pi}{2}} \frac{dx}{2+\sin x}$$

Onbepaald integraal: $\displaystyle\int \frac{dx}{2+\sin x}$

$$\tan\frac{x}{2} = t \Rightarrow x = 2\operatorname{Bgtan} t \Rightarrow dx = \frac{2dt}{1+t^2},\ \sin x = \frac{2t}{1+t^2}$$

$$\Rightarrow \int \frac{\frac{2}{1+t^2}}{2+\frac{2t}{1+t^2}}dt = \int \frac{\frac{2}{1+t^2}}{\frac{2+2t^2+2t}{1+t^2}}dt = \int \frac{2}{2+2t^2+2t}dt$$

$$= \int \frac{1}{t^2+t+1}dt = \int \frac{1}{t^2+t+\frac{1}{4}+\frac{3}{4}}dt$$

$$= \int \frac{1}{\left(t+\frac{1}{2}\right)^2+\frac{3}{4}}dt = \frac{2\sqrt{3}}{3}\operatorname{Bgtan}\frac{2t+1}{\sqrt{3}} + c = \frac{2\sqrt{3}}{3}\operatorname{Bgtan}\frac{2\tan\frac{x}{2}+1}{\sqrt{3}} + c$$

$$\Rightarrow \int_0^{\frac{\pi}{2}} \frac{dx}{2+\sin x} = \left[\frac{2\sqrt{3}}{3}\operatorname{Bgtan}\frac{2\tan\frac{x}{2}+1}{\sqrt{3}}\right]_0^{\frac{\pi}{2}} = \frac{\pi\sqrt{3}}{9}$$

References